

Who Gets Credit? Operationalizing AI Disclosure as Epistemic Coordination in Human-AI Teams

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Abstract

As generative AI becomes an ambient presence in collaborative work, a new social ambiguity emerges around authorship and responsibility. This condition of *authorship uncertainty* reshapes how teams attribute ideas, negotiate accountability, and coordinate collective reasoning. Prior research often treats AI presence as binary, framing it either as a hidden tool or a visible teammate. We argue that what matters in practice is the *design of disclosure*: how systems reveal, signal, or conceal AI assistance within collaboration. We introduce an *AI Disclosure Design Space* that conceptualizes disclosure as an epistemic coordination mechanism, articulating four configurations—No AI, Hidden AI, Translucent AI, and Visible AI—each trading off among accountability, autonomy, and coordination cost. We further contribute a research instrument that operationalizes these configurations in a collaborative chat setting and articulate testable design conjectures. By framing disclosure as epistemic infrastructure, this work outlines a conceptual roadmap for future empirical and design research on Human-AI collaboration.

CCS Concepts

• **Human-centered computing** → **Computer supported cooperative work**; **Collaborative interaction**; *HCI theory, concepts and models*; Empirical studies in collaborative and social computing.

Keywords

AI disclosure, transparency, human-AI teaming, teamwork, trust calibration, psychological safety, collective intelligence, AI-mediated communication

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1 Introduction and Related Work

Large language models (LLMs) are increasingly embedded in everyday teamwork, assisting with drafting messages, proposing ideas, and accelerating group problem-solving [2, 9]. Unlike traditional productivity tools such as spellcheckers, generative AI can produce

entire arguments, hypotheses, or strategies. As a result, when a strong idea appears in a shared workspace, teammates may no longer be certain whether it reflects a colleague’s reasoning or unseen AI assistance. We refer to this condition as *authorship uncertainty*.

Authorship uncertainty reshapes more than individual perceptions of effort; it alters how teams reason together. When the provenance of an idea is unclear, teammates may hesitate to challenge it, attribute disproportionate credit to its presenter, or defer judgment altogether. Prior work shows that AI assistance can reshape how expertise is perceived and distributed within teams [5], and that questions of credit and attribution become increasingly contested in human-AI co-creation [8]. Research on AI-mediated communication has further examined how transparency affects reliance and decision-making [10]. However, this line of work has largely focused on individual-level trust or adoption. It remains unclear how disclosure practices shape the collective processes through which ideas enter discussion, gain legitimacy, and stabilize within team reasoning.

We use the term *epistemic infrastructure* to refer to the shared rules, signals, and norms through which teams assess knowledge provenance and responsibility over time. In contrast, *epistemic coordination mechanisms* describe specific design interventions—such as AI disclosure—that actively shape how these norms operate in practice. This framing aligns with accounts of epistemic authority as socially negotiated and unevenly distributed, rather than purely merit-based [7].

Much of the human-AI teaming literature implicitly treats AI presence as binary. In many studies, AI appears either as a visible teammate whose outputs are shared with the group, or as an instrumental tool whose use remains outside the scope of analysis [6, 18]. In practice, however, AI often operates in an intermediate space. Individuals consult private assistants, selectively surface AI-informed ideas, and decide case-by-case whether to disclose assistance to others [19]. In these settings, visibility is not inherent to the technology but shaped by system design.

Different disclosure choices introduce distinct coordination problems. When AI use is hidden, teams may drift into forms of shadow AI, where ideas enter discussion without cues about their origin, increasing the risk of misattribution or suspicion [4]. Importantly, attribution processes are not neutral even in human-only teams: research documents systematic differences in how collaborative contributions are recognized, including gendered disparities in credit allocation [14], and evaluation contexts can exhibit bias that shapes perceived competence and legitimacy [12]. AI-mediated settings may therefore interact with, rather than replace, existing inequities in epistemic authority [7].



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When AI use is fully visible, transparency improves traceability but can also reshape participation norms. A shared AI agent may function as an epistemic shortcut that reduces human initiative, encourages automation bias, and increases coordination overhead [13, 17]. Related empirical work on human–AI collaboration in real organizational settings further suggests that AI integration can reconfigure knowledge work practices and accountability relationships beyond individual tool use [1]. Together, these trade-offs suggest that disclosure does more than signal tool use; it structures how teams generate, evaluate, and credit ideas over time.

This work shifts the focus from whether teams should use AI to how AI use should be disclosed within collaborative systems. We argue that disclosure functions as an *epistemic coordination mechanism* that shapes collective reasoning and contribution attribution. We introduce a design space of AI disclosure configurations for teamwork and present a working research platform that operationalizes these configurations. While this paper does not yet advance empirical claims, it introduces a fully implemented research platform and uses it to articulate testable conjectures grounded in concrete interactional affordances. Accordingly, this work asks: *How do different AI disclosure configurations function as epistemic coordination mechanisms that shape collective reasoning and attribution in collaborative teams?*

2 Research Instrument and Task Paradigm

This section operationalizes the AI disclosure design space through a controlled collaborative platform. Building on our view of AI disclosure as an epistemic coordination mechanism, we developed a web-based collaborative system to instantiate disclosure configurations for experimental study rather than to optimize productivity. The platform is fully implemented and functional. Its purpose is not to maximize AI performance, but to systematically manipulate disclosure conditions and observe how they shape collective reasoning, participation patterns, and contribution attribution under controlled conditions.

2.1 Design Rationale: Three Disclosure Tensions

Disclosure design introduces recurring tensions documented in prior research on coordination, automation, and human–AI interaction. We focus on three interrelated dimensions.

Accountability concerns whether teammates can infer the origin of ideas and assign responsibility for them. Research on epistemic authority and attribution demonstrates that contribution recognition is socially negotiated rather than mechanically assigned [7, 14]. In collaborative settings, visibility of provenance shapes how ideas are evaluated and credited.

Autonomy refers to individuals’ freedom to seek AI assistance without incurring social penalty, surveillance concerns, or strategic disadvantage. Studies of AI transparency and oversight suggest that mandatory visibility can alter user behavior and reduce perceived control [10].

Coordination cost captures the additional cognitive and interactional burden introduced by disclosure mechanisms, including signaling overhead, attention shifts, and conversational interruptions. Foundational work on coordination theory emphasizes that collaboration involves inherent coordination overhead [11], and

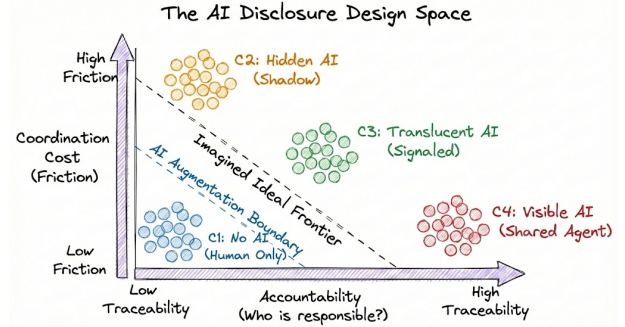


Figure 1: The AI disclosure design space as a conceptual landscape. The horizontal axis represents increasing accountability and traceability, while the vertical axis reflects coordination cost introduced by disclosure mechanisms. The dashed “imagined ideal frontier” illustrates a conceptual boundary where accountability is maximized without excessive coordination overhead; this frontier is heuristic rather than empirically established. The “AI Augmentation Boundary” represents the theoretical threshold where the coordination costs of managing AI disclosure begin to outweigh the collaborative benefits of the assistance.

automation research shows that added monitoring structures can reshape workload distribution [13].

We select these three dimensions because they repeatedly surface as structural tensions in human–AI collaboration research. They do not exhaust the space of possible objectives but provide analytically tractable axes for experimental manipulation.

2.2 Four Disclosure Configurations

We articulate four disclosure configurations spanning this design space, conceptually illustrated in Figure 1. These configurations were derived by synthesizing (1) research on hidden or “shadow” AI practices [4], (2) visible agent-based human–AI collaboration models [1], and (3) literature on automation bias and transparency effects [13, 17]. Together, they represent theoretically motivated extremes and intermediates along the axes of accountability, autonomy, and coordination cost.

These configurations differ in how disclosure signals are introduced and how epistemic authority is distributed within the team.

C1: No AI (Baseline). Teams rely solely on human effort, providing a reference point for patterns of idea generation, challenge, and attribution in the absence of AI mediation. This baseline does not assume that attribution is inherently fair; prior research documents persistent inequities in collaborative credit allocation [14]. Rather, it establishes a comparative condition for understanding how AI disclosure reshapes existing norms.

C2: Hidden AI (Shadow Configuration). Individuals may consult a private AI assistant without signaling this to teammates. Ideas may enter discussion without cues about how they were formed, concentrating epistemic authority at the individual level. This configuration maximizes autonomy and minimizes coordination overhead, but may heighten authorship uncertainty and complicate collective evaluation. In short-term or high-stakes collaboration,

such opacity may produce fragile attribution norms when outcomes are contested [8].

C3: Translucent AI (Signaled Configuration). Individuals may consult AI privately, while the system provides a lightweight signal indicating that AI assistance informed a contribution. In the current implementation, this signal takes the form of a binary “AI-assisted” tag attached to a message; however, translucency can vary in granularity in principle (e.g., signaling ideation versus drafting assistance). The AI’s raw output and interaction history remain private. We conceptualize this configuration as an epistemic prism: it signals provenance without exposing AI interaction artifacts (e.g., prompts and raw AI outputs). Contributors remain responsible for articulating and defending ideas within the shared workspace. Translucent disclosure therefore redistributes uncertainty rather than eliminating it, sustaining accountability while preserving human mediation.

C4: Visible AI (Agent Configuration). The AI appears as an explicit participant in the shared workspace. Queries and responses are visible to all team members, maximizing traceability and shared awareness. This configuration introduces a non-human epistemic actor whose outputs can anchor collective attention. Drawing on automation bias literature [17], visible AI may accelerate convergence while narrowing hypothesis exploration and redistributing responsibility for outcomes.

We focus on these four configurations as analytically tractable anchors that span key regions of the design space; they are not intended to be exhaustive, but to support controlled manipulation and comparison.

2.3 Platform Implementation

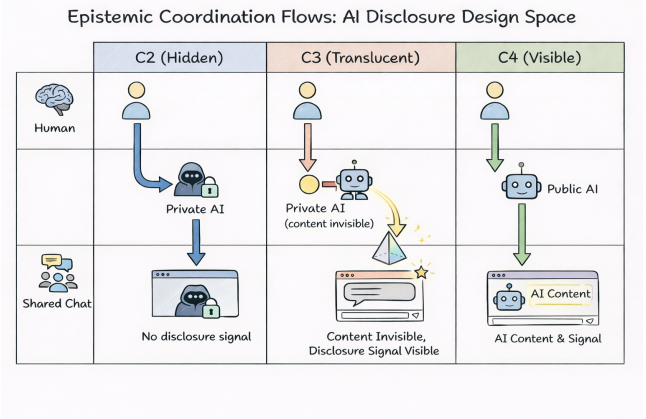
The platform supports real-time team text chat and allows disclosure conditions to be toggled while holding other interface elements constant. All teams interact in identical layouts; only disclosure signals differ. The system logs fine-grained interaction traces, including message timing, signal visibility, and AI query events.

Figure 2 illustrates both abstract interaction flows and concrete interface realization.

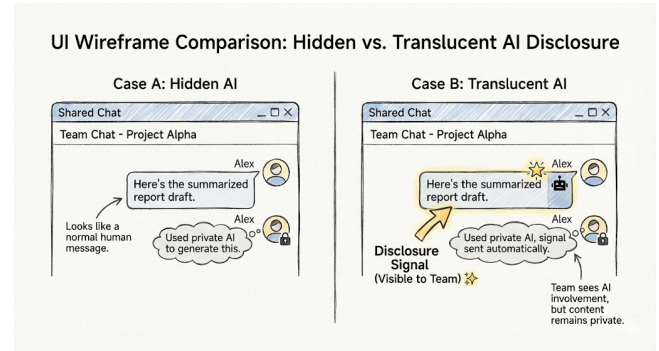
2.4 Task Paradigm

To foreground collective reasoning processes rather than outcome optimization, we pair the platform with lateral thinking puzzles as an epistemic task paradigm. These tasks require iterative hypothesis generation, questioning, and revision under shared uncertainty. While AI may support ideation or suggest lines of inquiry, it cannot trivially supply a complete solution without team coordination. Consequently, differences across disclosure configurations are expected to manifest primarily in interactional patterns—such as questioning frequency, challenge dynamics, and convergence trajectories—rather than simple task accuracy.

Building on these design choices, this operationalization grounds a set of theoretically motivated conjectures regarding how disclosure functions as an epistemic coordination mechanism. The next section articulates these conjectures.



(a) Abstract interaction flows under the four disclosure configurations, illustrating differences in provenance visibility and signal timing.



(b) Implemented chatroom interface showing the AI-assisted signal (Translucent condition) and visible AI participant (Agent condition).

Figure 2: Operationalizing AI disclosure in a collaborative chat setting: conceptual interaction models (top) and deployed interface conditions (bottom).

3 Discussion

3.1 Disclosure and Collective Reasoning Trajectories

As epistemic infrastructure, disclosure mechanisms reorganize how collective reasoning unfolds over time [15]. Team reasoning is a temporal process in which hypotheses are introduced, evaluated, revised, and either stabilized or discarded. Disclosure shapes this process by influencing how ideas enter the shared workspace and how they are interpreted by others.

When AI assistance is hidden, ideas may enter discussion without visible epistemic markers. Strong hypotheses can appear authoritative, leading teammates to attribute them to individual insight rather than algorithmic support. This dynamic may reduce visible critical questioning, particularly in early stages of problem solving when uncertainty is high. Over time, teams may exhibit earlier convergence patterns around ideas that appear compelling but whose underlying reasoning processes remain unexamined.

In such cases, the trajectory of collective reasoning may depend on opaque provenance signals rather than shared interrogation of evidence.

Fully visible AI produces a contrasting dynamic. When AI is a shared participant, its contributions are explicitly marked as non-human, improving traceability and reducing misattribution. At the same time, visible AI reorganizes the epistemic structure of the group. Human contributions may increasingly be evaluated relative to AI output rather than through peer-to-peer dialogue. Reasoning can shift from exploratory discussion among teammates toward consulting, filtering, or validating AI suggestions. Prior work on transparency effects suggests that such visibility can influence reliance patterns in collaborative decision-making [18]. While this configuration may accelerate convergence in some task contexts, it may also narrow the range of alternatives considered and alter patterns of productive disagreement.

Translucent disclosure introduces a distinct configuration. By signaling AI involvement without exposing content, it preserves ambiguity about internal reasoning paths while clarifying that assistance occurred. This ambiguity may prompt requests for justification and follow-up questioning, encouraging contributors to articulate and defend their reasoning rather than relying on perceived authority. In this sense, translucent disclosure may redistribute epistemic responsibility by sustaining both accountability and human mediation. Rather than eliminating uncertainty, it reshapes how uncertainty is negotiated within interaction.

3.2 Contribution, Credit, and Epistemic Authority

Disclosure mechanisms also shape how teams attribute contribution and negotiate epistemic authority. In collaborative work, credit is constructed through interaction, explanation, and recognition of effort over time rather than allocated solely on the basis of final outcomes. Authorship uncertainty complicates this process by blurring the boundary between human cognition and algorithmic assistance.

Hidden AI use concentrates epistemic authority at the individual level while distributing cognitive labor unevenly, particularly when only outcomes rather than reasoning processes are visible. Team members who consistently introduce strong ideas may accrue disproportionate credit, even when those ideas are AI-assisted. This can distort perceptions of expertise, especially in short-term collaborations where teammates lack longitudinal knowledge of one another's skills. If AI use is later revealed, prior attributions may be retroactively questioned, potentially destabilizing interpersonal trust and the legitimacy of earlier decisions [10]. This pattern aligns with prior observations that when visibility precedes verification, narratives and credibility assessments can stabilize before their provenance is clarified, making later corrections socially disruptive [16].

Visible AI redistributes authority by making algorithmic contributions explicit. While this can reduce ambiguity about provenance, it also raises questions about how human judgment is recognized. When AI outputs are public, the interpretive work of selecting, re-framing, or rejecting those outputs may become less salient within interaction logs. Over time, teams may treat AI suggestions as

neutral reference points, shifting evaluative attention away from human synthesis and altering how sensemaking labor is valued.

Translucent disclosure occupies an intermediate position in this negotiation. By acknowledging AI involvement while preserving human mediation, it frames contributors as interpreters rather than sole generators. This emphasizes responsibility for explanation, justification, and judgment as central forms of contribution in AI-augmented collaboration. At the same time, it raises open empirical questions about how teams recognize and reward interpretive labor relative to original proposal generation.

3.3 Disclosure as a Design Lever for Epistemic Equity

Disclosure functions not only as a transparency mechanism but as a structural lever that shapes epistemic equity within teams. Patterns of voice, challenge, and idea persistence are influenced by how AI involvement is signaled and interpreted.

Different collaborative contexts may warrant different disclosure configurations. In settings that prioritize speed or standardized outputs, fully visible AI may align with task objectives despite its impact on participation dynamics. In exploratory contexts, hidden AI may initially expand ideation but may also complicate longer-term attribution processes. Translucent disclosure constitutes a distinct configuration whose effects depend on how signals are designed and socially interpreted. Rather than advocating a universal disclosure policy, this framework treats disclosure as a configurable element of collaborative systems, adaptable to task demands and normative expectations.

Timing also plays a role. For example, disclosing AI assistance during early ideation may influence how teams explore alternatives, whereas disclosure introduced after convergence or decision-making may reshape how prior contributions are interpreted and evaluated [10].

3.4 Positioning Within the Broader Landscape

This framework complements emerging research on AI-enhanced collective intelligence and collaborative reasoning [3]. Work in this area emphasizes the interplay between human diversity, algorithmic support, and interaction patterns in sociotechnical systems. Our contribution extends this perspective by foregrounding disclosure as a structural interface between AI functionality and collective epistemic practice.

By framing disclosure as epistemic infrastructure, we offer a conceptual lens for examining how hybrid human–AI collectives generate knowledge, adjudicate competing proposals, and accumulate shared understanding over time. Empirical investigation is required to assess how these dynamics manifest across domains, task types, and institutional settings. The present framework provides a structured basis for such inquiry.

4 Conclusion and Future Directions

As generative AI becomes embedded in collaborative work, the central design question shifts from whether AI is used to how its use is disclosed within shared environments. Disclosure influences how ideas are interpreted, how legitimacy is established, and how responsibility and credit are negotiated among collaborators.

We frame this challenge as epistemic rather than purely technical, positioning disclosure as a system-level coordination problem in human-AI teamwork.

This paper introduced the concept of authorship uncertainty and articulated an AI disclosure design space that reframes disclosure not as a binary choice between secrecy and transparency, but as a configurable set of coordination mechanisms. By pairing this design space with a fully implemented research platform, we contribute both a conceptual framework and an operational instrument for systematically studying disclosure effects in collaborative settings.

The present work advances conjectures rather than empirical claims. Future research should examine how disclosure configurations shape collective reasoning, attribution dynamics, and participation patterns across varied collaborative contexts, including co-writing, planning, and longer-term teamwork. Empirical study is necessary to assess how the identified trade-offs among accountability, autonomy, and coordination cost manifest under different task conditions and institutional norms.

Rather than prescribing a universal disclosure policy, this work delineates the design space within which disclosure decisions can be evaluated. As human-AI collaboration continues to evolve, understanding how disclosure structures epistemic coordination remains an important direction for designing systems that support accountable and equitable collective work.

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